Ahmed Hassan

CS Student | Application Security & Secure Development

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Profile

Computer Science student at UET Lahore with hands-on experience in application security testing and secure software development. Skilled in identifying and analyzing OWASP Top 10 vulnerabilities using DAST tools (Burp Suite, OWASP ZAP) to understand root causes and remediation. Seeking to contribute to secure software lifecycles and vulnerability mitigation in a professional environment.

Technical Projects

SECURE SOFTWARE DEVELOPMENT PROJECTS | Full-Stack, Desktop, GUI Apps

Academic Projects | 2024 -- 2025

- Implemented secure coding practices, including input validation and parameterized queries in C#, to mitigate SQL injection vulnerabilities.
- Secured API keys and endpoints in a context-aware translation web app using environment variables and configuration management.
- Engineered algorithmic C++ applications optimizing data retrieval performance via hash maps and linked lists.

NETWORK SECURITY ANALYSIS & DEFENSE | Python, Web Protocols

Information Security Lab | Phase 1 Completed

- Simulated ARP spoofing and packet capture to analyze vulnerabilities in HTTP/HTTPS communication and evaluate strict HSTS configurations.
- Developed defense modules to establish network baselines, monitor IP-MAC mappings, and alert against anomalous activities.

ENTERPRISE NETWORK ARCHITECTURE | Cisco Packet Tracer

Academic Project | Spring 2025

- Engineered a multi-protocol enterprise network using RIP, EIGRP, and OSPF with route redistribution at HQ, enabling seamless communication across 4 retail stores. Implemented DHCP, dynamic NAT, and security controls for protecting critical assets such as POS systems and CCTV devices.

Cybersecurity Training

TRYHACKME & CTF LABS | Web Security & Mitigation

Self-Directed | 2024 -- Present

- Identified and analyzed OWASP Top 10 vulnerabilities (SQLi, XSS, IDOR) across 20+ web and network labs using Burp Suite and Nmap.
- Researched vulnerability root causes and mitigation techniques in sandbox environments to map attack surfaces and improve application security posture.

Education

UET LAHORE

B.Sc. Computer Science

Sep 2024 -- Present | 4th Sem | CGPA: 3.57

Skills

SECURITY TOOLS

Burp Suite (DAST) • OWASP ZAP (DAST)
Nessus • Nmap • Wireshark
Metasploit • Kali Linux • VirtualBox

NETWORKING

TCP/IP • DNS • HTTP/S • VLANs
NAT • DHCP • Firewalls • ACLs
Cisco Packet Tracer

PROGRAMMING

C/C++ • C# • Python • SQL
HTML5 • CSS • JS • Bash • LaTeX

SECURITY CONCEPTS

Application Security Testing
Secure SDLC Practices
Vuln. Assessment & Remediation
OWASP Top 10 • Threat Analysis
Privilege Escalation

Certifications

Completed: TryHackMe Pre-Security
In Progress: TryHackMe Jr. Pentester
Preparing for: CompTIA Security+, eJPT

Extracurriculars

Organizing Member | ITEC
Designed problem sets & logistics (100+ pax).

Participant | ITEC & ACM UET
Coding Competitions